Attendance Recovery Computation – Complete Python Solution

1. Problem Understanding

A student wants to calculate their current attendance percentage and determine how many additional consecutive classes they must attend to reach a required attendance eligibility percentage.

The program considers:

- Total classes conducted.
- Classes attended.
- Required attendance percentage.
- Future classes conducted.
- Future attendance scenarios.
- Minimum consecutive classes required to reach the eligibility threshold.

The program calculates current attendance, checks eligibility, and determines the minimum classes required under different future attendance scenarios.

Example: If a student has attended 60 classes out of 80, their current attendance is 75%. If the eligibility requirement is 75%, the student is already eligible.

2. Computational Thinking

Input:

- Student name or ID.
- Total classes conducted.
- Classes attended.
- Required attendance percentage.
- Number of future attendance scenarios.
- Future classes conducted in each scenario.
- Number of future classes the student plans to attend.

Processing:

1. Calculate current attendance percentage.

2. Check current eligibility.
3. Calculate the minimum additional consecutive classes required.
4. Evaluate different future attendance scenarios.
5. Calculate projected attendance percentages.
6. Display the eligibility status.

Output:

- Current attendance percentage.
- Current attendance status.
- Minimum consecutive classes required.
- Projected attendance under each scenario.
- Final eligibility result.

3. Important Formulas

Let:

- T = Total classes conducted.
- A = Classes attended.
- R = Required attendance percentage.
- x = Additional consecutive classes attended.

1. Current attendance percentage

$$\text{Attendance}=\frac{A}{T}\times100$$

2. Minimum consecutive classes required

If the student attends every upcoming class:

$$\frac{A+x}{T+x}\times100\geq R$$

Rearranging:

$$100(A+x)\geq R(T+x)$$

$$x(100-R)\geq RT-100A$$

Therefore, when the student is below the threshold:

$$ x=\left\lceil\frac{RT-100A}{100-R}\right\rceil $$

The program will calculate this using a loop rather than using a ceiling function.

3. Future attendance percentage

If F future classes are conducted and the student attends P of them:

$$ \text{Future Attendance}= \frac{A+P}{T+F}\times100 $$

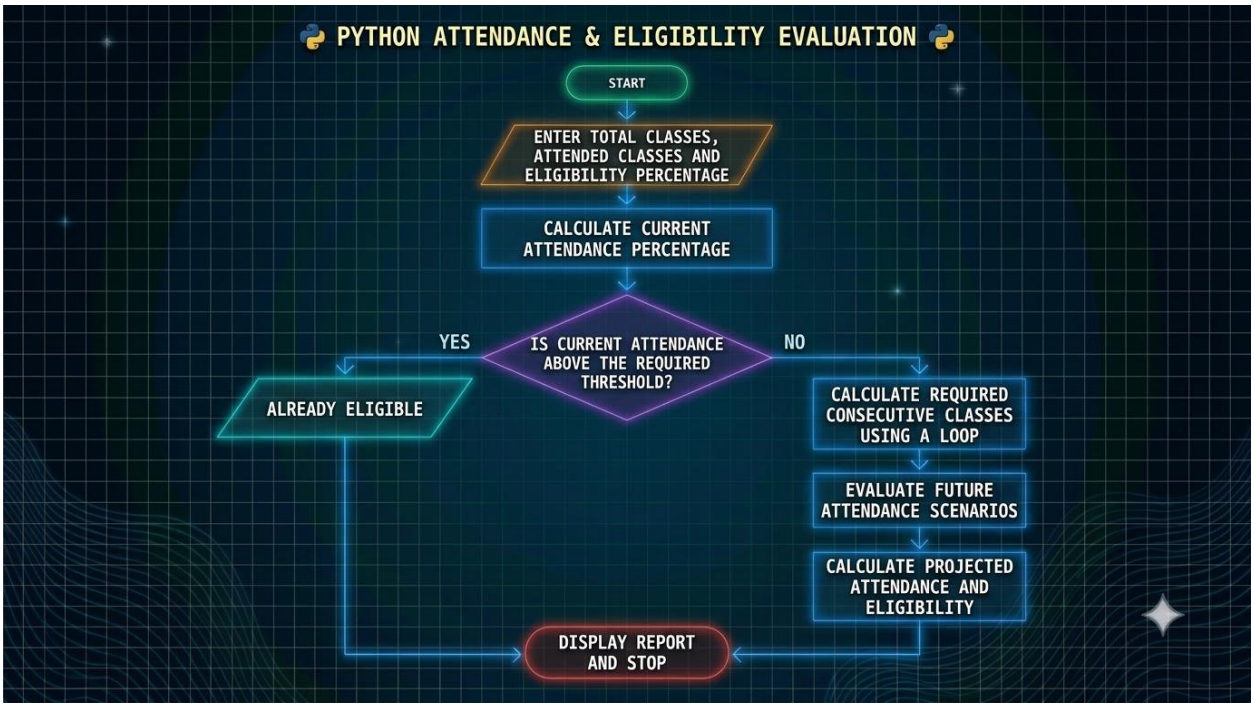
4. Algorithm

1. Start.
2. Enter student name.
3. Enter total classes conducted.
4. Enter classes attended.
5. Enter required attendance percentage.
6. Validate all inputs.
7. Calculate current attendance percentage.
8. Check whether the student is already eligible.
9. If not eligible:
    - Start additional classes at zero.
    - Increase the number of classes one by one.
    - Calculate projected attendance.
    - Stop when the required percentage is reached.
10. Enter the number of future scenarios.
11. For each scenario:
    - Enter future classes.
    - Enter classes expected to be attended.
    - Calculate projected attendance.
    - Check eligibility.

- Calculate the minimum consecutive classes needed if all remaining classes are attended.

12. Display all results.
13. Stop.

5. Flowchart



6. Pseudocode

START

INPUT student name

INPUT total classes

INPUT attended classes

INPUT required attendance percentage

IF total classes > 0 AND

attended classes <= total classes:

```
current_percentage =
    attended classes / total classes * 100

DISPLAY current percentage

SET additional_classes = 0

WHILE projected attendance < required percentage:

    additional_classes = additional_classes + 1

    projected attendance =
        (attended classes + additional_classes) /
        (total classes + additional_classes) * 100

DISPLAY additional classes required

INPUT number of future scenarios

FOR each scenario:

    INPUT future classes
    INPUT future attended classes

    projected attendance =
```

(attended + future attended) /

(total + future classes) * 100

CHECK eligibility

IF not eligible:

CALCULATE minimum consecutive classes

using a loop

DISPLAY scenario result

ELSE:

DISPLAY invalid input

STOP

7. **Complete Python Program**

```
# ATTENDANCE RECOVERY COMPUTATION

# 1. Input student details
student_name = input("Enter student name: ")

total_classes = int(input("Enter total classes conducted: "))

attended_classes = int(input("Enter classes attended: "))
```

```
required_percentage = float(
    input("Enter required attendance percentage: ")
)

# 2. Validate student details
if (total_classes <= 0 or attended_classes < 0 or
    attended_classes > total_classes or
    required_percentage <= 0 or required_percentage > 100):

    print("\nInvalid attendance details.")

else:

    # 3. Calculate current attendance
    current_percentage = (
        attended_classes / total_classes
    ) * 100

    print("\n====== CURRENT ATTENDANCE ======")

    print("Student Name:", student_name)
    print("Total Classes:", total_classes)
    print("Classes Attended:", attended_classes)
    print("Classes Absent:", total_classes - attended_classes)
```

```
print("Current Attendance:",
      round(current_percentage, 2), "%")

# 4. Find minimum consecutive classes required
additional_classes = 0

projected_percentage = current_percentage

if current_percentage >= required_percentage:

    print("Status: Already Eligible")
    print("Additional Classes Required: 0")

elif required_percentage == 100:

    print("Status: Not Eligible")
    print("100% attendance cannot be reached")
    print("because some classes were missed.")

else:

    while projected_percentage < required_percentage:

        additional_classes = additional_classes + 1

        projected_percentage = (
```

```
            (attended_classes + additional_classes) /
            (total_classes + additional_classes)
        ) * 100

    print("Status: Not Eligible")

    print("Minimum Consecutive Classes Required:",
          additional_classes)

    print("Attendance After Recovery:",
          round(projected_percentage, 2), "%")

# 5. Input future attendance scenarios
number_of_scenarios = int(
    input("\nEnter number of future scenarios: ")
)

if number_of_scenarios < 0:

    print("Invalid number of scenarios.")

else:

    # Nested list to store scenario details
    scenario_details = []
```

```
for i in range(number_of_scenarios):

    print("\nEnter Scenario", i + 1, "Details")

    future_classes = int(
        input("Enter future classes conducted: ")
    )

    future_attended = int(
        input("Enter future classes attended: ")
    )

    # Validate scenario
    if (future_classes < 0 or future_attended < 0 or
        future_attended > future_classes):

        print("Invalid scenario. Skipping scenario.")

        continue

    # 6. Calculate future attendance
    future_percentage = (
        (attended_classes + future_attended) /
        (total_classes + future_classes)
    ) * 100
```

```
# 7. Check eligibility
if future_percentage >= required_percentage:

    status = "Eligible"
    extra_classes = 0
    recovery_percentage = future_percentage

elif required_percentage == 100:

    status = "Not Eligible"
    extra_classes = -1
    recovery_percentage = future_percentage

else:

    status = "Not Eligible"

    # Find minimum additional consecutive classes
    extra_classes = 0

    recovery_percentage = future_percentage

    while recovery_percentage < required_percentage:

        extra_classes = extra_classes + 1
```

```
        recovery_percentage = (
            (attended_classes + future_attended +
             extra_classes) /
            (total_classes + future_classes +
             extra_classes)
        ) * 100

    # 8. Store details in nested list
    scenario = [
        "Scenario " + str(i + 1),
        future_classes,
        future_attended,
        future_percentage,
        status,
        extra_classes,
        recovery_percentage
    ]

    scenario_details.append(scenario)

# 9. Display scenario comparison
print("\n====== FUTURE SCENARIO REPORT ======")

for scenario in scenario_details:

    (name, future_classes, future_attended,
```

```
 percentage, status, extra_classes,
 recovery_percentage) = scenario

print("\n", name)

print("Future Classes:", future_classes)

print("Future Classes Attended:", future_attended)

print("Projected Attendance:",
      round(percentage, 2), "%")

print("Eligibility:", status)

if extra_classes == 0:

    print("Additional Classes Required: 0")

elif extra_classes == -1:

    print("100% attendance cannot be recovered.")

else:

    print("Additional Consecutive Classes Required:",
          extra_classes)
```

```
        print("Attendance After Recovery:",
              round(recovery_percentage, 2), "%")

# 10. Store student summary in tuple
student_summary = (
    student_name,
    total_classes,
    attended_classes,
    current_percentage,
    required_percentage
)

# 11. Tuple unpacking
(name, total, attended,
 percentage, target) = student_summary

print("\n====== FINAL SUMMARY ======")

print("Student:", name)
print("Total Classes:", total)
print("Attended Classes:", attended)

print("Current Attendance:",
      round(percentage, 2), "%")
```

print("Required Attendance:", target, "%")

8. Sample Input

Assume the student has the following attendance:

Enter student name: Student A

Enter total classes conducted: 100

Enter classes attended: 65

Enter required attendance percentage: 75

Future scenarios:

Scenario	Future classes	Classes attended
Scenario 1	20	20
Scenario 2	20	15
Scenario 3	30	30

9. Step-by-Step Calculations

Current attendance

$$\frac{65}{100}\times100=65\%$$

Required attendance = 75%.

The student is currently not eligible.

Minimum consecutive classes required

$$\frac{65+x}{100+x}\times100\geq75$$

$$6500+100x\geq7500+75x$$

$$25x\geq1000$$

$$x\geq40$$

40 consecutive classes

The student must attend 40 consecutive classes without missing any to reach 75%.

Scenario 1: Attend all 20 future classes

$$ \frac{65+20}{100+20}\times100 $$

$$ =\frac{85}{120}\times100=70.83\% $$

Status: Not Eligible.

Additional classes required after this scenario:

$$ \frac{85+x}{120+x}\times100\geq75 $$

$$ 25x\geq500 $$

$$ x=20 $$

Scenario 2: Attend 15 out of 20 classes

$$ \frac{65+15}{100+20}\times100 $$

$$ =\frac{80}{120}\times100=66.67\% $$

Status: Not Eligible.

Additional consecutive classes required:

$$ \frac{80+x}{120+x}\times100\geq75 $$

$$ 25x\geq1000 $$

$$ x=40 $$

Scenario 3: Attend all 30 future classes

$$ \frac{65+30}{100+30}\times100 $$

$$ =\frac{95}{130}\times100=73.08\% $$

Status: Not Eligible.

Additional consecutive classes required:

$$ \frac{95+x}{130+x}\times100\geq75 $$

$$ 25x\geq500 $$

$$ x=20 $$

10. Sample Output

====== CURRENT ATTENDANCE ======

Student Name: Student A

Total Classes: 100

Classes Attended: 65

Classes Absent: 35

Current Attendance: 65.0 %

Status: Not Eligible

Minimum Consecutive Classes Required: 40

Attendance After Recovery: 75.0 %

====== FUTURE SCENARIO REPORT ======

Scenario 1

Future Classes: 20

Future Classes Attended: 20

Projected Attendance: 70.83 %

Eligibility: Not Eligible

Additional Consecutive Classes Required: 20

Attendance After Recovery: 75.0 %

Scenario 2

Future Classes: 20

Future Classes Attended: 15

Projected Attendance: 66.67 %

Eligibility: Not Eligible

Additional Consecutive Classes Required: 40

Attendance After Recovery: 75.0 %

Scenario 3

Future Classes: 30

Future Classes Attended: 30

Projected Attendance: 73.08 %

Eligibility: Not Eligible

Additional Consecutive Classes Required: 20

Attendance After Recovery: 75.0 %

====== FINAL SUMMARY ======

Student: Student A

Total Classes: 100

Attended Classes: 65

Current Attendance: 65.0 %

Required Attendance: 75.0 %

11. Additional Test Cases

Test case	Total classes	Attended	Required	Expected result
1	100	65	75%	40 additional classes
2	100	75	75%	Already eligible
3	80	60	75%	Already eligible
4	100	50	75%	100 additional classes
5	100	90	75%	Already eligible

12. Assigned Syllabus Concepts Used

Syllabus topic	Application
Computational thinking	Breaking attendance calculations into steps
Algorithms	Finding the required classes
Flowcharts	Representing program flow
Pseudocode	Planning program logic
Variables	total_classes, attended_classes
Data types	Integer, float, string
Operators	Arithmetic and comparison
Conditional statements	if, else
Loops	for and while
Strings	Student name and scenario names
Pattern generation	Not required
Lists	Storing future scenarios
Nested lists	Storing multiple scenario details
Tuples	Storing student summary
Tuple operations	Storing and accessing summary data
Unpacking	Extracting tuple values into variables

Important assumptions:

- Each future class attended increases both the attended-class count and total-class count by one.
- Future scenario classes are considered completed before calculating the projected percentage.
- Additional recovery classes are assumed to be attended consecutively, without absence.
- The program uses 75% as an example threshold. Actual attendance requirements should be checked against the student's college regulations.

Conclusion: The program calculates current attendance and finds the minimum consecutive classes required to reach the specified eligibility percentage under different future attendance scenarios.